

ATPL EASA STUDY GUIDE | 021 - AIRFRAME & SYSTEMS

Chapter 18: Aviation Gasoline, Carburation & Lubrication

Syllabus Source: CAE Oxford (pp. 49-92 (Pwr)) | Pass Target: >= 90% AviationExam

1. AVIATION GASOLINE (AVGAS): GRADES, COLOR CODES & DETONATION

Piston aircraft burn Aviation Gasoline (AVGAS) engineered for high volatility resistance and anti-knock stability:

Fuel Grade	Identification Dye Color	Octane / Performance Rating & Typical Use
AVGAS 100LL (Low Lead)	DYED BLUE	Most common aviation gasoline worldwide. 100 octane rating; contains limited tetraethyl lead (TEL) anti-knock additive. Density ~0.72 kg/litre.
AVGAS 100 (Green)	DYED GREEN	High-lead 100/130 aviation gasoline. Used in high-power supercharged vintage piston engines.
AVGAS 80/87	DYED RED	Low-octane fuel for vintage low-compression light aircraft engines.
Jet A-1 (Kerosene)	CLEAR / STRAW (Colorless)	STRICT PROHIBITION: NEVER PUT JET FUEL IN A PISTON ENGINE! Causes immediate catastrophic engine detonation, complete power loss, and engine destruction on take-off!

- **Detonation (Knocking):** Uncontrolled explosive auto-ignition of the remaining unburned end-gas ahead of the normal flame front. Creates supersonic shockwaves, extreme cylinder head temperatures (CHT), and can burn holes through piston crowns within seconds! Caused by: low fuel octane, excessively lean mixture, high MAP with low RPM, or high CHT.
- **Pre-Ignition:** Combustion initiated BEFORE the spark plug fires, caused by an incandescent hotspot in the cylinder (glowing carbon deposit, cracked spark plug insulator). Can lead directly to violent detonation.

2. CARBURETTOR ICING: TYPES & AMBIENT CONDITIONS

Float-type carburetors are highly susceptible to ice formation even in warm, sunny weather:

- **Fuel Evaporation Ice:** Liquid gasoline vaporizing in the venturi absorbs latent heat of vaporization, dropping air temperature inside the carburetor throat by up to 20°C to 25°C!
- **Throttle Ice:** Pressure drop across a partially closed throttle butterfly valve causes adiabatic expansion cooling, causing moisture to freeze directly onto the throttle valve.
- **Icing Risk Envelope:** Carburetor icing can occur with Outside Air Temperature (OAT) as high as +25°C or even +30°C if relative humidity is above 80%! Most severe risk occurs between -7°C and +21°C at moderate-to-high humidity and low power settings (descent/glide).
- **Carb Heat Operation:** Selects unfiltered air ducted across the hot exhaust shroud. When carb heat is selected ON: 1. Manifold pressure / RPM drops initially (hot air is less dense); 2. Engine runs rough as melted ice passes through; 3. RPM / MAP then rises higher than original as ice clears.

[!] EXAM TRAP (AVIATIONEXAM) - Carburettor Heat Application Sequence Trap

- Carburettor heat is an ANTI-ICING and DE-ICING control. When selected ON, it admits hot, UNFILTERED air into the engine. - Ground Rule: Do NOT use carb heat on the ground in dusty conditions (unfiltered air sucks grit into cylinders) except for brief pre-take-off functional check! - Throttle Rule: Carb heat should ALWAYS be applied BEFORE reducing power for descent, because exhaust shrouds cool rapidly at low throttle!

3. ENGINE LUBRICATION SYSTEMS: WET VS DRY SUMP

Engine oil cools internal components, lubricates bearing journals to reduce friction, cleans away carbon particles, cushions impact loads, and forms a gas seal between piston rings and cylinder walls:

- Wet-Sump System: Oil is stored in the crankcase oil pan (sump) beneath the engine. Used on light aircraft (Lycoming O-320/O-360). Simple, but oil sloshes during aerobatics/maneuvers, risking pump cavitation.
- Dry-Sump System: Oil is stored in an external separate oil tank. A scavenge pump (with capacity 25-50% larger than the pressure pump) draws oil from the shallow crankcase pan, routes it through an oil cooler, and pumps it back to the tank.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Leaning to Peak EGT & Carburettor Icing**

1. Leaning Protocol: Slowly pull mixture control -> EGT rises to maximum (Peak EGT)
-> Advance mixture slightly to '50 F Rich of Peak' for best engine longevity. 2.
Carburettor Icing on Constant-Speed Propeller: - Governor holds RPM constant. -
Ice restriction is detected exclusively as an UNEXPLAINED DROP IN MANIFOLD PRESSURE (MAP). - Apply FULL CARB HEAT immediately. Expect engine roughness before power recovers.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What ambient weather conditions present the highest risk for severe carburettor icing in an aircraft piston engine?**

- [A]: OAT -20°C with 10% relative humidity
- [B]: OAT between -7°C and +21°C with high relative humidity (>80%), especially at low power settings
- [C]: OAT +35°C in dry desert air
- [D]: Only when flying through freezing rain

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Fuel vaporization drops air temperature by up to 25°C. When ambient air is between -7°C and +21°C with high humidity, this temperature drop freezes moisture rapidly, choking the carburettor venturi.

Q2: What is the aviation dye color code for AVGAS 100LL (Low Lead)?

- [A]: Green
- [B]: Red
- [C]: Blue

- [D]: Amber

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. AVGAS 100LL is dyed BLUE worldwide. (AVGAS 100 is green, AVGAS 80 is red, and Jet A-1 is clear/straw).

Q3: Why does engine RPM drop slightly when carburettor heat is selected ON in an aircraft with a fixed-pitch propeller?

- [A]: The ignition timing retards
- [B]: Hot air is less dense than cold ambient air, reducing charge mass entering the cylinders and producing a richer, less powerful mixture
- [C]: The throttle valve closes partially
- [D]: The exhaust valve sticks open

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Hot air ducted from the exhaust shroud has lower density, which reduces the mass of air entering the engine, slightly richening the mixture and causing an immediate small drop in power/RPM.